

Which fusion is worth testing?

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Team Fusion turns scattered chimeric-RNA evidence into a source-linked next experiment.

THE PROBLEM

A ranked gene pair is not yet an exact RNA junction—or a functional protein.

OUR APPROACH

Trace each claim to reads, probes, structures and the paper. Let the evidence determine the next check.

RNA, junction, protein: separate the proof

Reported RNA support is a starting point. Each next claim needs its own evidence.

01 RNA PAIR

**Do these genes
appear together?**

109 reported
NanoString-supported pairs

02 EXACT JUNCTION

**Where do the two
sequences join?**

Read alignments +
probe coordinates

03 PROTEIN

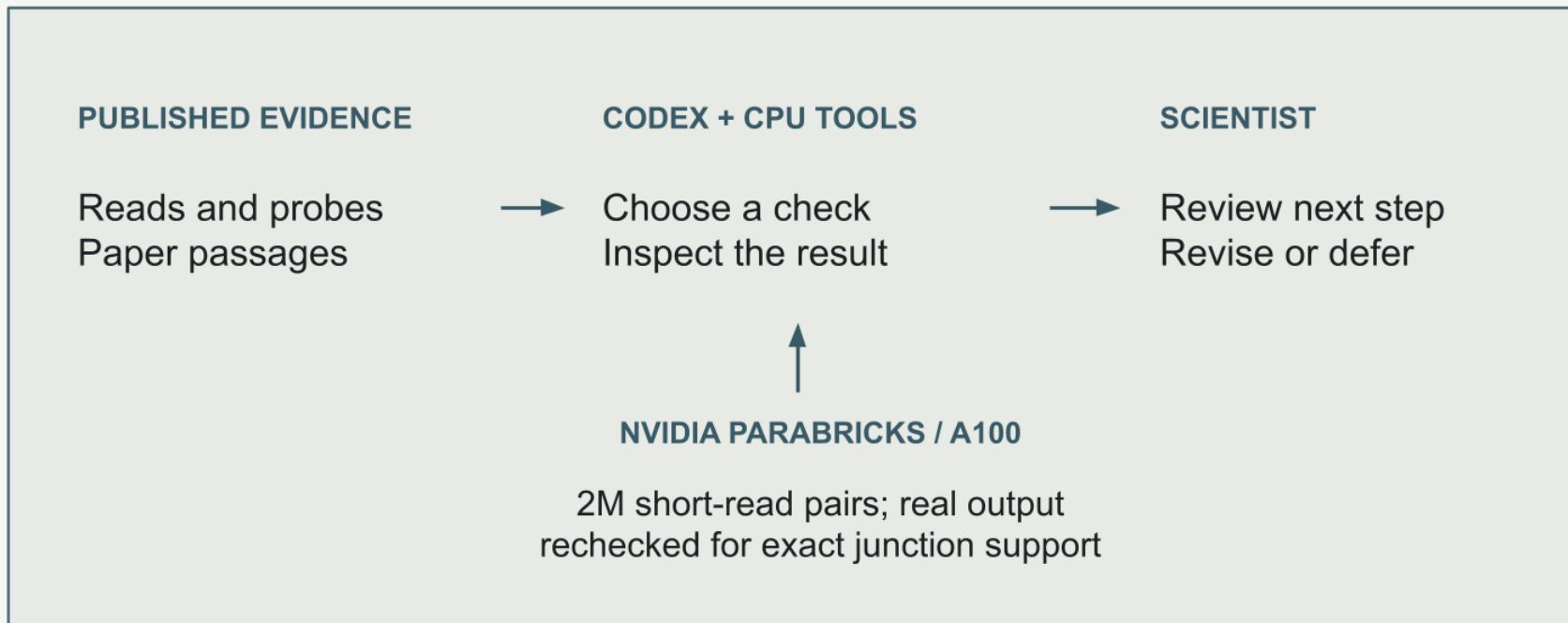
**Is it translated
and functional?**

Protein and functional
experiments

One evidence layer cannot stand in for the next.

From candidate to scientist review

Codex executed 12 bounded tool calls across three cases on Brev. Every recorded output replays exactly.



924 bases change the next question

Psap-Lgals3

924 bases

Genomic endpoint discrepancy



Nearest read
47,622,058

Probe
47,622,982

EVIDENCE

Six published read IDs.
The nearest endpoint still differs.

NEXT ACTION

Inspect the source alignment
before selecting the junction
assay.

WHAT THIS MEANS

A discrepancy to resolve;
not proof of false RNA.

Direct source audit: Supplementary Tables 3/8 + GENCODE M28.
Descriptive discrepancy; long-read coordinate convention is unresolved.

A model is one layer of evidence



Gsdmd–Tmem106a · reference hypothesis

PUBLISHED EVIDENCE

Gsdmd–Tmem106a has protein and functional experiments in the paper.

OUR MODEL

118-aa reference hypothesis.
Mean pLDDT 48.7.
Low confidence.

INTERPRETATION

Model uncertainty does not refute the experiments.

Model: Boltz2 2.2.1. Colours show parent segments, not confidence.
Experimental context: Venezia et al., Nature (2026), Figs 3–4.

Does Hi-C improve the ranking?

Held-out average precision

RNA features

0.296

RNA + Hi-C

0.300

0

1

MATCHED EVALUATION

401 pairs with complete Hi-C.
Five gene-disjoint folds.

UNCERTAINTY

Δ AP: +0.004
95% interval:
-0.031 to +0.047

CONCLUSION

No established gain.
Assay support is the label.

Same 401-pair cohort; interval resamples gene components.
Unreported NanoString support is not a verified negative.

Make the next experiment defensible.

TEAM FUSION Source-linked evidence review for chimeric RNA

A visible chain from the scientific claim to the evidence—and the next decision.

DELIVERED

- 3 reviewed cases
- 12 exact tool replays
- 30 passing tests
- Public reproducible release

NEXT VALIDATION

- Resolve source alignments.
- Test with independent scientists.
- Measure whether review improves decisions.

github.com/cheparukhin/team-fusion-hackathon

Technical appendix

PROJECT LINKS

GitHub: [team-fusion-hackathon](#)

Offline: [demo/review/index.html](#)

Reviewed science: 76fb5de

Public repository; see release manifest

HOW TO REPRODUCE

Python 3.12 · uv · requirements.lock

python scripts/reproduce.py --download

python scripts/review/verify.py

Demo: python scripts/demo/serve.py

Core rebuild: identical predictions

TECHNICAL DETAILS

NVIDIA: Parabricks 4.7.1-1

A100 80 GB · 2M read pairs

85.49 s alignment; no speedup claim

OpenAI: Codex / gpt-6-astra

Recorded tool use; offline replay

Data: [Venezia et al., Nature \(2026\)](#)

GRCm39 / GENCODE M28

479 pairs; 5 gene-disjoint folds

Hi-C: AP 0.296 → 0.300 (n=401)

Δ AP 95% CI: -0.031 to +0.047

Python packages pinned; CPU replay

Cached Boltz2 2.2.1: low confidence